Northern University of Business and Technology Khulna

Department of Computer Science and Engineering

Fall CS25: Introduction to C Programming

Fall 2025 — Course Syllabus

Contents

Important Link

For the full topic list and additional resources, visit:

Course Topic List

Teacher Advisor

Vashkar Kar
Lecturer of CSE
Northern University of Business and Technology Khulna.

Course Overview

This course introduces core programming concepts using C. Students will learn fundamental programming constructs, problem-solving techniques, and hands-on coding practices. Sessions are delivered twice weekly.

Course Modules (Short List)

1. What is Programming?
2. Intro to C and Setting up the Environment
3. Data Types, Variables and Input/Output
4. Operators
5. Conditional Statements
6. Loops
7. Functions
8. Arrays
9. Strings
10. Pointers and References
11. Structures

Content Table

Weekly Plan

- **Class Days:** Thursday and Sunday
- **Time:** 8:00 PM (local)

Weekly Modules and Key Topics

No.	Module Title	Key Topics / Outcomes
1	What is Programming?	Problem solving, algorithms, pseudo-code, overview of programming languages.
2	Intro to C and Environment	Installation of compiler/IDE, first program, compilation process, code structure.
3	Data Types, Variables, I/O	Primitive types, variables, constants, formatting output.
4	Operators	Arithmetic, relational, logical, bitwise operators, operator precedence.
5	Conditional Statements	`if / else-if / else`, nested conditions, `switch`.
6	Loops	`for`, `while`, `do-while`, break/continue, common loop patterns.
7	Functions	Declaration, definition, parameters, return types, variable scope.
8	Arrays	1D and 2D arrays, iteration, common array algorithms.
9	Strings	C-style strings, string operations, parsing, concatenation.
10	Pointers and References	Memory addressing, pointer arithmetic, references, dynamic memory basics.
11	Structures	`struct` definitions, nested structures, arrays of structures, practical examples.

Program Instructors & Assistants

- **Program Instructors**
  - Mir Mashrafi Hossain Sahil — Program Instructor
  - SK. Tasnim UR Rahman — Program Instructor
  - Sabbir Hossain — Program Instructor
- **Program Assistants**
  - Devjyoti Tikader — Program Assistant

Program Organizers

- Rahul Kumar Ghosh — Program Coordinator
- Saiful Islam — Program Coordinator
- Sourav Mondal — Program Coordinator
- Ayon Adhikary Arko — Program Coordinator

Rules and Regulations

The following rules and regulations apply to all students enrolled in this course:

1. Students must submit all assignments and projects by the given deadlines. Late submissions may be penalized.
2. Academic integrity must be maintained. Any form of cheating or plagiarism will result in disciplinary action.
3. Active participation in lectures and practice sessions is encouraged.
4. Students are expected to maintain a respectful and professional attitude towards instructors and peers.
5. Online etiquette must be observed: inappropriate language, disruptive behavior, or sharing offensive content in the online classroom or chat will not be tolerated.
6. Any student found violating these rules may be removed from the session or reported to program authorities for further action.